

FULL-MIX HISTORICAL MUSIC RESTORATION IN LATENT SPACE

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ABSTRACT

Previous literature on historical music restoration has almost exclusively focused on constrained problems such as super-resolution or single-instrument material, leaving the task of restoring multi-instrument historical music largely unexplored. This is especially challenging for early-20th-century symphonic recordings because models must learn combinations of unknown nonlinear historical degradations while identifying and restoring the high-fidelity characteristics of multiple instrument combinations, all without available paired data. This paper introduces the full-mix historical music restoration problem to the audio machine-learning community and proposes a predictive latent-processing configuration that subjectively and objectively outperforms existing state-of-the-art historical restoration baselines. Furthermore, we curate a license-free, unpaired historical classical-music test set and introduce a historical degradation function whose distributional proximity, measured using MAUVE Audio Divergence, shows substantially better simulation of historical distortions than previously used degradation models. Finally, through objective non-intrusive and intrusive evaluation metrics, we provide compelling evidence that restoration performance depends more on modeling of historical degradation than on methodological complexity. Our code and demo is available at the official repository.¹

Index Terms— music restoration, SAME-L, conditional flow matching, transformers, latent diffusion

1. INTRODUCTION

In 1877, Thomas Edison invented the phonograph, giving the world, for the first time, the ability to record music [1]. Then in only 100 years, recording technology rapidly advanced to almost lossless recordings. Still, many of the time’s legendary artists such as Fats Waller, Tchaikovsky, and Debussy recorded a vast amount of their discography with lossy and noisy recording equipment such as wax cylinders, 78-rpm vinyl disks, gramophones, etc. [2], leaving much of their valuable works corrupted and underappreciated. This makes the task such as historical music restoration valuable with abundant applications.

Historical music restoration began with heuristic digital signal processing methods, including blind deconvolution for unknown channel degradations [3] and statistical restoration methods for clicks and hiss suppression [4]. Eventually, neural systems trained on synthetically corrupted music showed that learned models can improve historical-music [5]. Soon after, many methodologies began addressing music and audio restoration such as super resolution (SR). SR methods for example, to recover higher frequency information lost to an ideal lowpass filter on high-fidelity audio [6], have tried training generative methods such as diffusion [7], flow

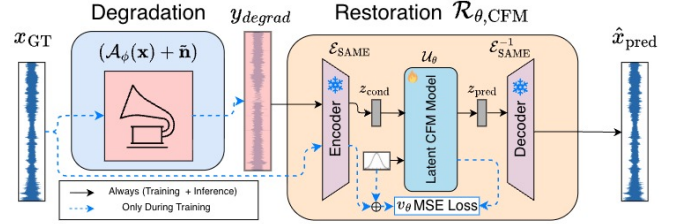


Fig. 1: Overview of the proposed historical music restoration pipeline $\mathcal{R}_{\theta,CFM}$.

matching [8], or Schrödinger Bridges [9] on paired waveform domains [10] and frequency domain representations such as Complex STFT [11] and Mel-Spectrograms [12]. While, the waveform and spectrogram space was adequate for training speech restoration models, training a generalizable restoration model on music was much more difficult due to its larger source variety of sound [?, 4]. Therefore, newer models began applying predictive methodologies onto latent spaces of waveforms and spectrograms to learn on more compact and expressive features [6, 13, 14].

While this significantly improved music SR models’ generalization [14], because historical music degradations never exclusively involve ideal lowpass filters, the SR models still fail to restore historical music due to its complex non-linear distortions and spectral attenuations [15]. As a result, some methods opt to restore in the unpaired blind inverse problem space in the frequency domain, blindly inferring the degradation of the audio while simultaneously restoring it [15, 16]. Still, the methodology has only been proven for single instrument stems such as piano, vocals, and strings [16].

We propose a new method that trains transformers [17] on synthetically degraded paired Stable Audio 3 [18] latent embeddings that is capable of restoring both on single and multi-stem music recordings authentic or synthetic. The synthetic degrader involves a 5-stage nonlinear distortion that sufficiently models the degradations and distortions of historical music recorded in the early 20th century. We chose to use synthetic paired dataset due to the benefits outweighing the cons of paired restoration training.

For this paper, we define music to be historical when it was published before 1925. Electrical recording was popularized at 1925 [19], and recording quality rapidly improved after. 1925 also falls within the public domain for free use. This enables us to release all paper related datasets and demos [20].

To introduce the problem space of multi-instrument historical music restoration, we make the following contributions:

- We introduce a new **149-song test set** of real 78-rpm historical classical recordings for evaluating restoration under authentic degradations.
- We propose a **synthetic degradation model** based on a Wiener-

¹<https://github.com/stevencho-sony/>

Hammerstein structure to approximate the distortions characteristic of 78-rpm historical recordings.

- We apply a **Latent-space Conditional Flow Matching (CFM) restoration model** on the full-mix historical music restoration space that substantially outperforms prior state-of-the-art baselines.

2. METHODS

2.1. Problem Definition

Let $\mathbf{y} \in \mathbb{R}^T$ denote an observed full-mix historical music recording of duration T samples. We assume this recording originates from an unknown, clean, high-fidelity recording of the same duration, denoted $\mathbf{x} \in \mathbb{R}^T$. The relationship between the two is given by the forward model

$$\mathbf{y} = \mathcal{A}(\mathbf{x}) + \mathbf{n}, \quad (1)$$

where $\mathbf{n}$ represents common additive degradations in historical recordings, such as hiss and clicks [4], and $\mathcal{A}$ denotes the composition of non-additive degradations, including bandwidth limitation, spectral coloration, nonlinear distortion, and other unknown degradation processes.

Historical music restoration therefore seeks to recover an estimate $\hat{\mathbf{x}} \approx \mathbf{x}$ of the unknown high-fidelity recording from $\mathbf{y}$ alone. Critically, both $\mathcal{A}$ and $\mathbf{n}$ are unknown, making historical music restoration a blind inverse problem: the degradation process must be inverted without any explicit knowledge of what needs to be corrected. This blindness renders the problem severely ill-posed: infinitely many pairs $(\mathbf{x}, \mathcal{A})$ could explain the same observation $\mathbf{y}$. Moreover, even if $\mathcal{A}$ were known, the problem would remain ill-posed, since degradations such as bandwidth limitation are inherently non-invertible and irrecoverably discard information about $\mathbf{x}$. Recovering $\hat{\mathbf{x}}$ therefore requires both a strong prior over what plausible high-fidelity music looks like and a model of the degradation process itself.

2.2. Supervised-learning approach

We address this problem by learning a restoration model $\mathcal{R}_\theta : \mathbf{y} \mapsto \hat{\mathbf{x}}$, parametrized by θ , that approximates the inverse of the degradation process. We adopt a supervised-learning paradigm, which has proven effective across a wide range of blind inverse problems in audio and image restoration [11, 21], by learning $\mathcal{R}_\theta$ directly from paired examples of degraded and clean signals rather than by explicitly modeling and inverting $\mathcal{A}$ and $\mathbf{n}$.

This requires training data in the form of clean/distorted pairs $(\mathbf{x}, \mathbf{y})$, from which θ is optimized to minimize a loss between $\mathcal{R}_\theta(\mathbf{y})$ and $\mathbf{x}$. Such paired data is, however, not available for real historical recordings: only the degraded observation $\mathbf{y}$ exists, with no access to a corresponding clean ground truth $\mathbf{x}$.

We propose training with synthetic pairs $(\mathbf{x}, \mathcal{A}_\phi(\mathbf{x}) + \tilde{\mathbf{n}})$, where clean examples $\mathbf{x}$ are drawn from a dataset of high-fidelity full-mix music, $\tilde{\mathbf{n}}$ represents isolated additive disturbances drawn from a noise dataset [22], and $\mathcal{A}_\phi$ is a purpose-designed synthetic degradation model, described next.

2.3. Synthetic degradation model

We require a degradation model $\mathcal{A}_\phi$ whose parameters ϕ can be randomly sampled during training. Rather than attempting to match the true distribution of real degradations $p_{\mathcal{A}}$, we design $\mathcal{A}_\phi$ together with a parameter distribution p_ϕ such that the induced distribution over operators is broad enough to encompass $p_{\mathcal{A}}$. Training on examples $\mathcal{A}_\phi(\mathbf{x})$ with $\phi \sim p_\phi$ then exposes the restoration model $\mathcal{R}_\theta$ to a

Table 1: MAD Scores for Full-Mix + Sections Across Progressive Degradation Conditions. Scores are computed against the historical unpaired dataset using MERT, Fx-Encoder++, and SAME-L embeddings. **Bold** and underlining indicate the best and second-best scores, respectively, within each column. The final row directly compares the two IA subsets.

Degradation	MAD Score ($\downarrow$)					
	MERT		Fx-Encoder++		SAME-L	
	Orch.	Light Orch.	Orch.	Light Orch.	Orch.	Light Orch.
Clean	5.39	5.40	5.43	5.49	5.33	5.45
Gaussian Noise	5.18	5.13	4.71	5.08	5.40	5.44
Low-Pass + Gaussian Noise	5.31	5.46	5.42	5.49	5.24	5.37
Gramophone Noise	3.90	4.42	3.86	4.63	<u>4.57</u>	4.82
WH + Gramophone Noise (Ours)	<u>4.23</u>	<u>4.55</u>	<u>4.42</u>	<u>4.99</u>	4.54	<u>4.95</u>
Orch. vs. Light Orch.	2.08		2.67		3.00	

wider family of degradations than it will encounter at inference, encouraging generalization to real recordings $\mathbf{y} = \mathcal{A}(\mathbf{x}) + \mathbf{n}$ without requiring $\mathcal{A}$ to be identified.

Accordingly, we design $\mathcal{A}_\phi$ to simulate a large family of degradations, covering not only bandwidth limitation and spectral coloration as in previous work [11, 16], but also nonlinear system behaviour. We build it around a Wiener–Hammerstein (WH) model, a simple yet expressive block-oriented structure classically used for nonlinear system identification [23], consisting of a static nonlinearity Ψ placed between two linear filters H_{pre} and H_{post} . This structure provides a compact approximation of historical recording and playback chains, in which frequency-dependent coloration interacts with amplitude-dependent distortion. The complete model is given by

$$\mathcal{A}_\phi = H_{\text{BPF}} \circ H_{\text{post}} \circ \Psi_{\text{nonlin}} \circ H_{\text{pre}}. \quad (2)$$

The linear blocks H_{pre} and H_{post} are zero-phase finite impulse response (FIR) filters realized in the frequency domain, similar to [24, 25]. Their magnitude responses are parameterized by a set of attenuation weights (ϕ_{pre} and ϕ_{post}), one per octave band, which are linearly interpolated, while the phase response is constrained to zero. An additional bandpass filter H_{BPF} complements H_{post} and explicitly models the restricted bandwidth typical of historical recordings. It is likewise a zero-phase FIR filter implemented in the frequency domain, but follows the parameterization of [16], comprising low and high cutoff frequencies (f_{low} and f_{high}) together with two roll-off slopes (s_{low} and s_{high}) that define the attenuation rate beyond each cutoff.

The nonlinear block Ψ_{nonlin} is a static nonlinearity, implemented as a scaled tanh [24] with a wet/dry residual connection,

$$\Psi_{\text{nonlin}}(\mathbf{x}) = (1 - w)\mathbf{x} + w \tanh(a\mathbf{x})/a, \quad (3)$$

with parameters $\phi_{\text{nonlin}} = \{w, a\}$. The additive degradation $\tilde{\mathbf{n}}$ signal-to-noise ratio is set by ϕ_{add} .

The full parameter set $\phi = \{\phi_{\text{pre}}, \phi_{\text{post}}, \phi_{\text{BPF}}, \phi_{\text{add}}\}$ collects the coefficients of all four blocks and is resampled independently for every training example. The choice of p_ϕ is critical: distributions that are too narrow risk teaching $\mathcal{R}_\theta$ to invert one particular degradation, while deliberately broad distributions favor robustness to unknown historical conditions at the cost of some realism. We designed p_ϕ by analyzing a dataset of historical recordings and through qualitative listening, and report the resulting parameters in the official repository². We validated our degradation model by computing

²<https://github.com/stevencho-sony/>

the MAUVE Audio Divergence (MAD) score between the synthetically degraded output with internal collection of internet archive historical classical recordings, and our degradation model significantly improved other common methods as shown in table 1. The Gramophone Noise method is ablated and analyzed in section 3.5 and 4

2.4. Restoration Model

We implement the latent restoration model $\mathcal{R}_{\theta, \text{CFM}}$ to solve the ill-posed inverse problem $\mathcal{R}_{\theta} : \mathbf{y} \mapsto \hat{\mathbf{x}}$. Latent Processing (LP) has demonstrated promising results in audio SR [6, 13], therefore we employ a variational autoencoder (VAE) $\mathcal{E} : \mathbb{R}^T \rightarrow \mathbb{R}^{C \times N}$ to embed the historical recordings into latent vectors $\mathbf{z}_y \in \mathbb{R}^{C \times N}$ where C and N are the number of channels and frames respectively.

We then implement latent conditional flow matching $\mathcal{R}_{\theta, \text{CFM}}$ inspired by Liu et al.’s LatentFlowSR [13]. Unlike LatentFlowSR, $\mathcal{R}_{\theta, \text{CFM}}$ does not have additional spectral and semantic embeddings, and we use SAME-L for our VAE and transformers as our backbone. CFM’s training and inference, is based on Lipman et al.’s Flow matching configuration [8]. The objective is $\mathcal{L}_{\text{CFM}} = \mathbb{E}_{\mathbf{z}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), t \sim \mathcal{U}(0, 1)} \|v_{\theta}(\mathbf{z}_t, t, \mathbf{z}_{\text{cond}}) - \mathbf{v}\|_2^2$. During training, a time-conditioned transformer learns the velocity from the Gaussian source toward the clean target with drop out probability $p_{\text{uncond}} = 0.1$. During inference, $\mathcal{A}_{\phi}(\mathbf{x}) + \tilde{\mathbf{n}}$ is used to create the conditioning $\mathbf{z}_{\text{cond}}$, and the learned velocity field is integrated using the Euler ODE solver.

3. EXPERIMENTAL SETUP

3.1. VAE and Backbone Architecture

For the encoder $\mathcal{E}$ and decoder $\mathcal{E}^{-1}$, we use SAME-L [18] the autoencoder used in Stable Audio 3. We validated our choice for SAME-L by comparing reconstruction error with other SOTA encoders/decoders. This reduces the restoration problem from $\mathcal{R}_{\theta} : \mathbf{y} \mapsto \mathbf{x}$ to $\mathbf{z}_y \mapsto \mathbf{z}_x$.

The backbone for $\mathcal{U}_{\theta, \text{CFM}}$ is approximately 40M parameters with transformer architecture: 768-dimensional tokens, 9 AdaLN-Zero blocks, 12 attention heads, and an MLP width of 1536.

3.2. Training and Validation Data

We restrict our historical music restoration to classical music. Due to abundance of high quality public classical music datasets and genre-inherent consistency of piece renditions, this minimizes genre-related out of distribution concerns when generating degraded-clean paired training sets to model real historical recordings. The paper will also not address music with vocals as vocal restoration is an independent restoration field and lies outside of this paper’s scope.

To construct the training data pairs $(\mathbf{x}, \mathcal{A}_{\phi}(\mathbf{x}) + \tilde{\mathbf{n}})$, we construct a public instrumental-classical corpus from PHENICX-Anechoic [26], ChoraleBricks [27], URMP [28], ORCHSET [29], and the Beethoven Symphony Dataset (BSD) [30] for $\mathbf{x}$. The resulting training corpus *full mix + sections* (FMS) contains 5 h 10 min of full-mix music, with 1 h 39 min additional brass, woodwind, and string section mixtures from PHENICX-Anechoic, ChoraleBricks, and URMP, totaling approximately 6 h 49 min of classical music data: full mixes and section mixes account for 75.8% and 24.2%, respectively. All audio is pre-processed to mono, resampled to 44.1 kHz, and standardized to -23 LUFS.

Section mixtures are included to increase available training data while directly exposing instrument-family specific timbre and high-frequency structure to the model. No isolated percussion sections were available. Finally, FMS corpus is segmented into 5-s windows using a 10% window hopsize.

For the validation set, validation windows are randomly assigned with overlap-aware data leakage prevention: once a window is assigned for validation, all windows overlapping that window is discarded. Random validation window assignments occur until the validation window to remaining windows ratio surpasses the 9:1 train-validate ratio. For each training and validation window, 20 iterations of $\mathcal{A}(\mathbf{x}) + \mathbf{n}$ is applied, encoded, and saved along with the ground truth for efficient training. The synthetic degradation model is applied only after the train-validation partitioning to ensure differently degraded versions of the same window cannot leak into the training set.

3.3. Test Data

We curate a paired test set and an unpaired historical test set. Because there are no established metrics to measure restoration quality for unpaired historical music, we construct the paired test set to compute restoration scores on both established reference-based metrics and unpaired repurposed audio quality metric to validate, along with subjective scores, its credibility when analyzing restoration on unpaired historical recordings.

The paired set is derived from the Spheres classical music dataset (53.0 min) [31]. We only use the provided full-mix recordings. All audio is preprocessed like the training set except the recordings are divided into non-overlapping 5s windows.

The unpaired historical music test set (9.85 h) consists of digitized 78-rpm recordings collected from Internet Archive and is exclusively instrumental orchestral classical music published in the United States between 1911 and 1925. It is also up-sampled to 44.1kHz FLAC files, converted into mono, and noise normalized to -23 LUFS. In addition, it is sorted into two groups: Orchestra, consisting of symphonic orchestral recordings, and Light Orchestra, consisting of wind-ensemble and chamber recordings. This will serve to analyze the importance of sub-genre train-test domain matching for historical restoration. The official data can be found on Zenodo³

3.4. Objective Evaluation

We assess $\mathcal{R}_{\theta, \text{CFM}}$ on the paired test set with Mel-Spectrogram Loss (Mel-MSE) [32], CLAP [33] cosine similarity (CosSim), AA-PQ [34], and MAD [35]. The four metrics jointly cover *spectral fidelity* and *content preservation* – both essential when restoring historical recordings with minimal hallucinations and high production quality. The method restores every degraded window, and the metrics score and average with equal weight from the window scores to one song score and the song scores to one test set score - which is reported to the table 2. If needed, outputs are resampled before scoring (e.g. 22.05 kHz BEHM-GAN) for equal comparisons with methods operating at 44.1 kHz. For historical unpaired test set, only reference free metrics were calculated.

Unpaired Metric: AA-PQ (Audiobox Aesthetic Production Quality) [34] is an established reference-free objective metric of perceived production quality sensitive to fidelity, frequencies, and dynamics.

Paired Metrics: MEL-MSE is the error between mel-spectrograms of the restored and reference audio, providing a perceptually weighted measure of tonal and timbral reconstruction quality. CLAP CosSim measures a pair’s similarity in CLAP’s musical/semantic embedding space of instrumentation, musical character, timbral identity, and broad acoustic characteristics at 48kHz. Finally, MAD

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Table 2: Objective and Subjective Restoration Performance Using Full-Mix + Sections Training. Objective metrics are reported on historical unpaired and synthetic paired test sets. CLAP cosine similarity measures paired similarity in CLAP embedding space, while MAD compares MERT embedding distributions. AA-PQ measures no-reference production quality. Subjective results report MOS-Preservation and MOS-Quality on a 1–5 scale; boxes show the mean and 95% confidence intervals, while whiskers show min/max. **Bold** and underlining indicate the best and second-best available model results, respectively.

Method	Params	Historical Unpaired Test Set				Synthetic Paired Test Set			
		AA-PQ ($\uparrow$)		MOS-P ($\uparrow$)	MOS-Q ($\uparrow$)	AA-PQ ($\uparrow$)	Mel MSE ($\downarrow$)	CLAP Cos. ($\uparrow$)	MAD ($\downarrow$)
		Orch.	Light Orch.						
Unprocessed input	–	4.94	5.00	–	1.06	5.74	3.94	0.44	5.03
Ground truth	–	–	–	–	4.44	7.44	0.00	1.00	0.00
BEHM-GAN	82M	4.87	4.93	–	–	6.05	5.33	0.45	4.94
BABE2	40M	5.88	5.87	3.44	<u>3.47</u>	6.35	4.99	0.56	5.46
Denoiser	73M	5.74	6.02	–	–	6.35	5.27	0.51	1.21
BEHM-GAN-FMS	82M	5.67	5.66	–	–	6.09	<u>1.97</u>	0.65	4.09
BABE2-FMS	40M	5.54	5.41	3.97	2.69	6.24	3.22	0.65	<u>1.82</u>
Denoiser-FMS	73M	5.95	<u>6.17</u>	–	–	6.88	2.25	<u>0.71</u>	4.19
SAMECFM	40M	6.70	6.86	<u>4.47</u>	3.78	7.21	1.90	0.73	3.70
only Gram. Degrad.	40M	<u>6.07</u>	6.06	4.50	3.31	6.08	3.92	0.54	5.10

$= -\log(\text{MAUVE})$ [35] measures the divergence of the restored test audio distribution with the ground truth test audio distribution in the MERT embedding space at 24kHz.

Taken together, these metrics measure the overall perceptual/production and musical quality and help validate the reference-free metrics in the full-mix historical restoration domain.

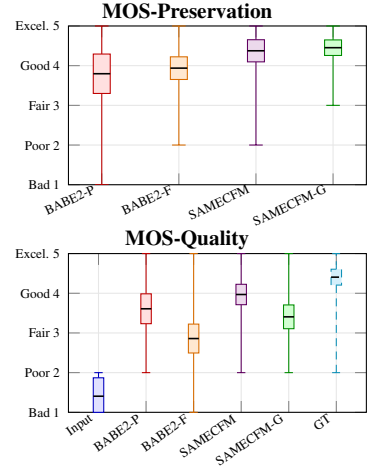
3.5. Subjective Evaluation

For subjective evaluation, we conduct two mean opinion score (MOS: MOS-Q, MOS-P) tests on restored historical music. MOS-Q evaluates overall audio quality across historical input, high-quality reference, BABE2 pretrained (BABE2-P), BABE2 FMS (BABE2-F), SAMECFM $\mathcal{R}_{\theta, \text{CFM}}$, and its gramophone ablation (SAMECFM-G) $\mathcal{R}_{\theta, \text{CFM-G}}$. The difference between $\mathcal{R}_{\theta, \text{CFM}}$ and $\mathcal{R}_{\theta, \text{CFM-G}}$ is that the paired FMS set for the latter only degrades with the gramophone stage **n**. Twenty-five listeners were asked, “Rate the overall audio quality between 1 and 5, considering noise, timbre, sound clarity, and high-frequency details”. MOS-P evaluates how well four restoration methods – BABE2-P, BABE2-F, SAMECFM, and SAMECFM-G – preserve the historical input. Listeners were asked, “Rate how well the reference audio retains the essential features of the source audio, such as melody, instrumentation, and overall style. Use 1 (Poor Preservation) to 5 (Excellent Preservation).” Both tests use ten 5-s excerpts equally divided between Orchestra and Light Orchestra recordings, with each method receiving 100 ratings. We report mean scores with 95% confidence intervals. We removed a total of N testers due to failing to rate the high anchor ground truth with an average score of at least 4.

4. RESULTS

We compare pretrained and FMS-domain-adapted BEHM-GAN, BABE2, and Denoiser against our proposed $\mathcal{R}_{\theta, \text{CFM}}$. Domain adaptation uses the FMS dataset together with our proposed historical degradation model, while $\mathcal{R}_{\theta, \text{CFM-G}}$, trained using only Gramophone noise degradation, serves as an ablation.

As shown in Table 2, the pretrained baselines transfer inconsistently to full-mix historical music, often improving one aspect of restoration while degrading another. FMS adaptation generally improves performance, particularly from paired to unpaired than light



orchestra to orchestra, demonstrating the importance of matching the degradation process to the target historical domain over matching style. However, the magnitude of improvement remain dependent on the underlying restoration method.

The subjective evaluation provides the clearest assessment of this trade-off. Among the subjectively evaluated methods, preservation of the original recording is generally strong, with $\mathcal{R}_{\theta, \text{CFM}}$ receiving the highest MOS-Preservation score and $\mathcal{R}_{\theta, \text{CFM-G}}$ following closely. BABE2-P is the main exception: although listeners rate its restored audio relatively highly in MOS-Quality, its substantially lower preservation score suggests that this quality gain is achieved partly by altering characteristics of the original recording. FMS adaptation improves BABE2 preservation, but at the cost of a lower perceived quality score.

Overall, $\mathcal{R}_{\theta, \text{CFM}}$ provides the strongest balance between restoration quality and preservation, achieving the highest MOS-Quality among restoration methods while also obtaining the highest MOS-Preservation score. Its weaker MAD relative to the best baselines suggests a modest trade-off in distribution-level similarity for improved perceptual quality. Notably, this degradation is not reflected in the human preservation ratings, indicating that MAD and subjective preservation capture complementary aspects of restoration fidelity. Finally, the consistent advantage of the full model over $\mathcal{R}_{\theta, \text{CFM-G}}$ across objective and subjective evaluation further demonstrates the importance of modeling the broader combination of historical degradations.

5. CONCLUSION

This paper introduces Full-Mix Historical Music Restoration as a 44.1kHz blind inverse problem extending beyond conventional super-resolution and single-instrument restoration. Across objective and human evaluations, domain adaptation using full-mix and section-level training data, together with a broader nonlinear historical degradation model, substantially improves restoration quality. Our latent CFM model achieves the strongest overall perceptual and paired reconstruction performance while maintaining strong preservation of the historical input. Its consistent improvement demonstrates the importance of accurately modeling the broader historical degradation process. The results also reveal a quality–preservation trade-off that is not fully captured by any single objective metric.

Future work will refine degradation simulation, extend restoration beyond instrumental classical music, and develop metrics that jointly measure perceptual quality and content preservation.

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